

Intro to Factoring & Factoring with GCF Notes

Warm Up	Directions: Simplify the following polynomials: • $a(3a + 7)$ = _____ • $-2m(m^2 + 6m - 1)$ = _____ • $4x^3y(x^2 - 2y)$ = _____
	(Simplest Form) ➔ (Factored Form) Polynomials that cannot be factored are called _____!

Factoring a GCF (greatest common factor)	There are several factoring methods; the approach depends on the polynomial. We will start by identify and factoring out the greatest common factor (GCF) of the polynomial.
	Steps for Factoring GCF: Step 1: Identify the GCF of the polynomial: • Check the coefficients for a GCF. • Now look at the variables. A variable must be present in all terms to be a GCF. If a variable I present in all terms, take the one with the smallest exponent. Step 2: Divide each term by the GCF and leave the remaining factors in parentheses Step 3: Check your work by distributing.

Directions: Factor each polynomial. Check your work by distributing. If a polynomial cannot be factored, write "prime"

1. $3x + 12$

2. $7y - 7$

1. $6a^2 + 27$	2. $4y^2 - 24y$
3. $21cd - 3d$	4. $14gh - 18h$
5. $ab - a$	6. $m^3n - m^2n^2 + 5mn^3$
7. $5x - 13$	8. $35a^2 - 20ab^2 + 15a$
9. $2x^2y - 2xy^2 + 4xy$	10. $16xy^2 + 28xy + 8y$